



National and Kapodistrian University of Athens (NKUA)

MSc in Data Science and Information Technologies (DSIT)

Course: *Molecular Modeling of Biomolecules (2025–2026)*

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Mid-Term Project

1.

KRAS encodes a small GTPase that acts as an on/off molecular switch in cell signaling. It alternates between an inactive GDP-bound state and an active GTP-bound state, thereby controlling major pathways involved in cell growth, proliferation, and survival. KRAS4B is anchored to the inner plasma membrane, which is essential for its signaling function.

KRAS is one of the most frequently mutated oncogenes in human cancer, with activating mutations such as G12D leading to continuous signaling due to defective GTP hydrolysis. Because these mutations drive uncontrolled cell proliferation and are found in a large fraction of tumors, studying the biology of KRAS is crucial for understanding cancer development and for guiding the development of new therapeutic strategies, especially since effective drugs for many KRAS mutants are still lacking.

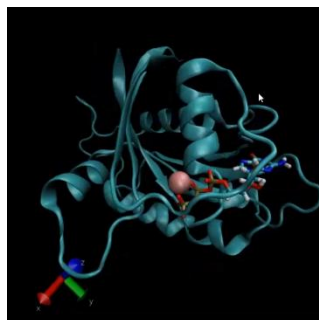


Figure 1. Representation of the KRAS-G12D protein bound to GTP and Mg^{2+} . The GTP molecule is shown in the active site together with the magnesium ion, depicted as a spherical bead. The protein backbone of KRAS is shown in blue, highlighting the overall fold of the G-domain surrounding the nucleotide-binding pocket.

2.

The G12D KRAS mutant has the typical KRAS protein structure, dominated by a conserved catalytic G-domain that binds GDP or GTP. This domain is composed of α -helices and β -sheets and contains the nucleotide-binding site that controls the active and inactive states of the protein. Two regions within the G-domain, known as Switch I and Switch II, change conformation depending on whether GDP or GTP is bound and are important for interaction with downstream signaling proteins.

The G12D mutation is located in the phosphate-binding loop of the G-domain and does not significantly alter the overall fold of the protein. However, it locally affects the nucleotide-binding region and disrupts normal GTP hydrolysis, favoring the active state of KRAS. As a result, the main structural framework remains intact, while subtle local changes explain the abnormal signaling behavior of the mutant.

Protein preparation, PSF generation, and energy minimization were carried out using VMD and NAMD.

First, the KRAS G12D PDB structure was inspected in VMD and preprocessed by removing non-essential crystallographic components (e.g. waters or ions not required for the initial setup). Missing hydrogen atoms were added, and protonation states were assigned according to physiological pH. The structure was checked for missing residues or atoms to ensure compatibility with the force field.

Next, the PSF file was generated using the AutoPSF/psfgen plugin in VMD. CHARMM topology files were loaded, protein segments were defined, and no patches were applied. This step produced a consistent PSF file describing the molecular topology and atomic coordinates of the system.

The protein was then solvated in an explicit water box using the VMD solvate plugin, ensuring sufficient padding between the protein and the box boundaries. System neutrality and physiological ionic strength were achieved by adding counterions using the autoionize plugin.

Finally, energy minimization was performed in NAMD under periodic boundary conditions using the CHARMM force field and Particle Mesh Ewald (PME) electrostatics. Convergence was assessed by monitoring the total energy decrease, ensuring a **stable** starting structure for subsequent MD simulations.

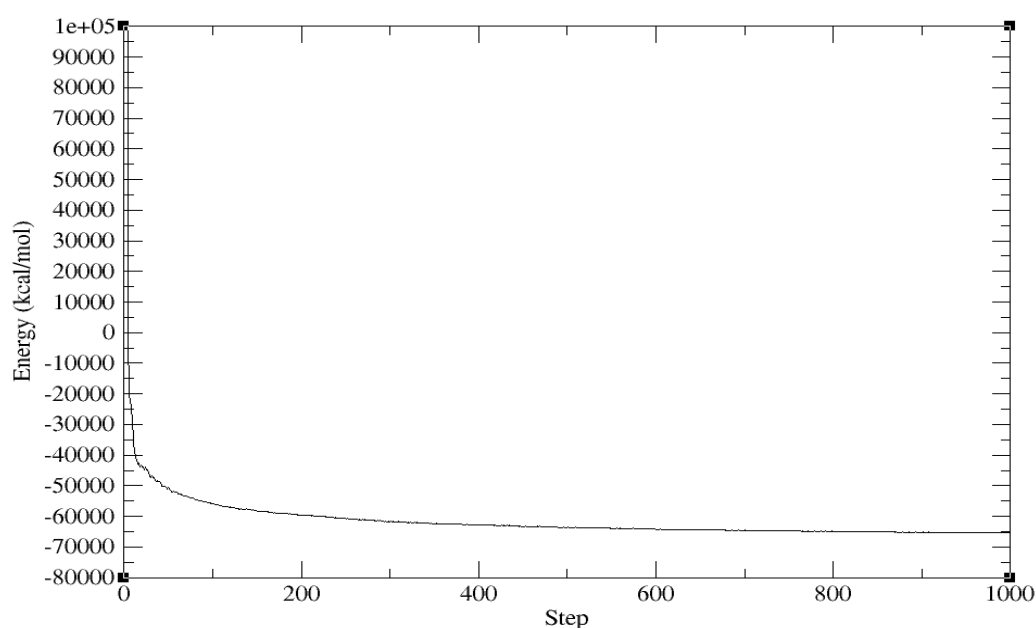


Figure 2. Potential energy (kcal/mol) versus minimization step for the KRAS-G12D protein in explicit solvent. The rapid energy decrease followed by a plateau indicates successful convergence of the energy minimization prior to molecular dynamics simulations.

3.

The KRAS-G12D system was prepared and simulated following the standard molecular dynamics workflow described in the tutorial. After energy minimization, the system was gradually heated from

low temperature to 310 K under constant volume and temperature (NVT) conditions. The heating stage was performed in small temperature increments, and each temperature step was restarted from the restart files generated in the previous step (coordinates, velocities, and extended system), ensuring a continuous and physically consistent evolution of the system.

Following heating, an equilibration phase was carried out, again restarting from the final restart files of the heating stage. The equilibration was performed in the **NVT ensemble**, maintaining a constant temperature of 310 K while keeping the simulation box dimensions fixed. The equilibration was performed in the NVT ensemble at a constant temperature of 310 K with fixed simulation box dimensions. Although the tutorial proposes an NPT equilibration using a Langevin piston, attempts to apply pressure coupling resulted in numerical instabilities and repeated errors related to periodic cell fluctuations, despite appropriate parameter adjustments. To ensure a stable and physically meaningful equilibration, NVT dynamics were therefore employed, allowing the system to relax energetically after heating without introducing additional volume degrees of freedom.

Throughout the equilibration, the total energy of the system was monitored as a function of simulation time. The energy decreases initially and then reaches a stable plateau, indicating that the system has relaxed and reached a steady state. This behavior confirms that the heating and equilibration protocols were successful and that the structure is well equilibrated at the target temperature.

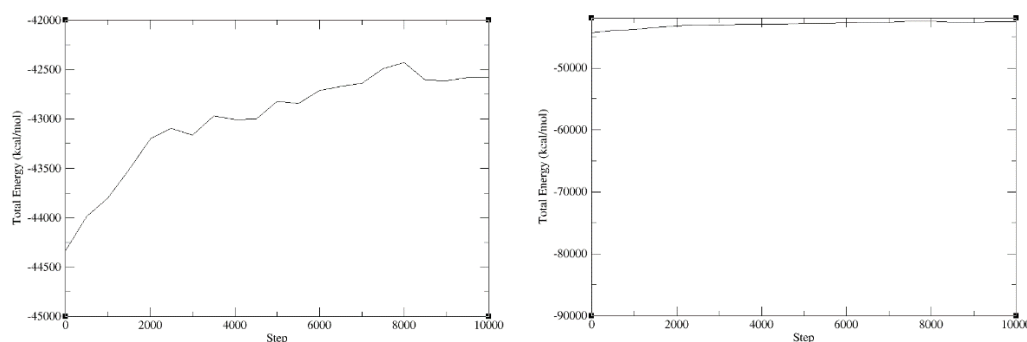


Figure 3. Total energy of the KRAS-G12D system as a function of simulation time during the equilibration phase at 310 K (NVT ensemble). The left panel shows a zoomed-in energy scale highlighting the relaxation of the system toward equilibrium, while the right panel uses a broader energy scale to emphasize the overall stability and steady-state behavior achieved during equilibration.

4.

Based on the analysis of the potential energy as a function of simulation time, the protein **has reached equilibrium**. After an initial relaxation phase, the potential energy decreases and subsequently converges to a well-defined plateau, exhibiting only small thermal fluctuations around a constant mean value. This behavior is characteristic of an equilibrated system and indicates that no systematic energy drift is present.

Therefore, additional equilibration cycles are **not required**. The protein can be considered energetically equilibrated under the conditions of our simulation.

5.

A production molecular dynamics simulation of KRAS G12D was performed for 1 ns under NVT conditions. System stability during the production run was verified by monitoring the total energy and temperature as a function of time, which remained stable throughout the simulation.

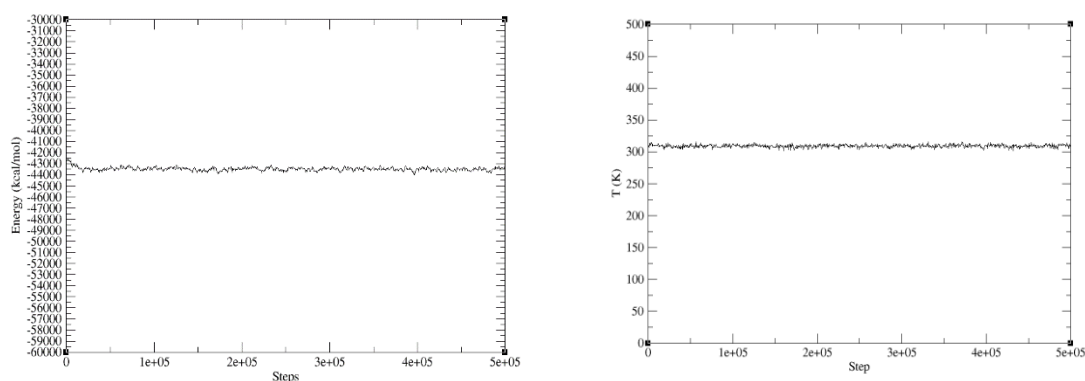


Figure 5. Total energy (left) and temperature (right) of the KRAS G12D system as a function of simulation steps during the production molecular dynamics run. Both quantities fluctuate around stable mean values, indicating that the system remains thermodynamically stable and well equilibrated throughout the production phase.

6.

a) The potential salt bridge between ASP154 and ARG161 involves side-chain atoms rather than the backbone. Specifically, the negatively charged carboxylate oxygens (OD1 and OD2) of ASP154 and the positively charged guanidinium nitrogens (NH1 and NH2) of ARG161 are the atoms that could, in principle, participate in a salt bridge. However, analysis of the time series of the ASP154–ARG161 interatomic distance shows that the separation remains consistently above the typical cutoff for salt bridge formation ($\approx 3.5\text{--}4.0\text{ \AA}$) throughout the trajectory. Therefore, no stable or transient salt bridge is formed during the simulation. The observed distance fluctuations reflect general thermal motion and conformational flexibility rather than repeated formation and disruption of a salt bridge. This behavior is consistent with residues that are spatially separated or solvent-exposed under the existing conditions.

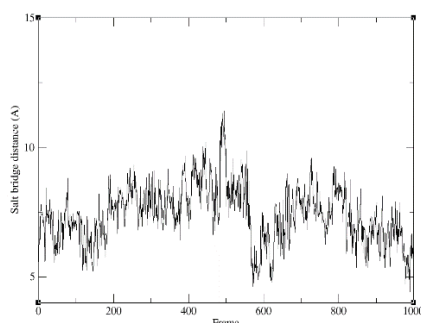


Figure 6a. Time series of the salt bridge distance between ASP154 and ARG161 during the production MD simulation. The plot shows the distance between the carboxylate oxygen atom of ASP154 (OD1) and the guanidinium nitrogen atoms of ARG161 (NH1) as a function of simulation time, illustrating the transient formation and disruption of the salt bridge over time.

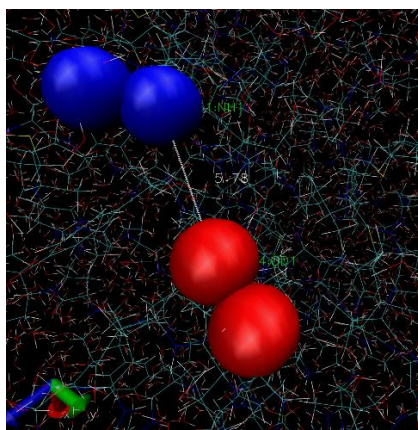


Figure 6b. Representative snapshot from the production MD trajectory illustrating the salt bridge between ASP154 and ARG161. The interaction is formed between the negatively charged carboxylate oxygen atoms of ASP154 (OD1/OD2, shown in red) and the positively charged guanidinium nitrogen atoms of ARG161 (NH1/NH2, shown in blue). The dashed line indicates the instantaneous interatomic distance used to monitor the salt bridge along the trajectory.

b) It is attached (includes 2 videos)



c)

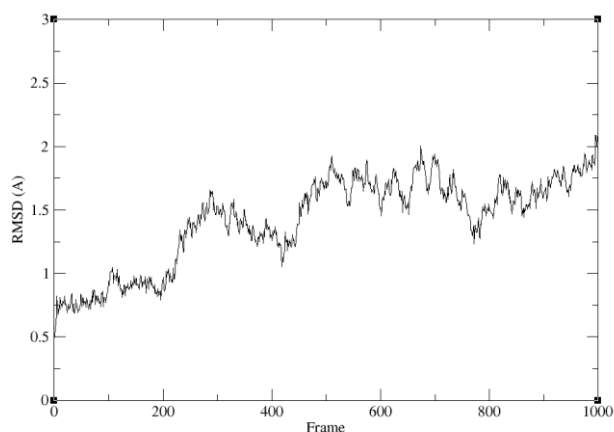


Figure 6c. Root-mean-square deviation (RMSD) of the KRAS protein backbone ($C\alpha$ atoms) as a function of simulation frame during the production molecular dynamics run. The RMSD is calculated relative to the initial reference structure (frame 0), illustrating the conformational stability and dynamic fluctuations of KRAS over time.

The RMSD time series of KRAS was computed using the $C\alpha$ atoms of the protein backbone, with the initial structure (frame 0) as the reference. The RMSD shows an initial increase during the early part of the trajectory, reflecting structural relaxation from the starting conformation. Following this equilibration phase, the RMSD fluctuates around a stable plateau.

Quantitatively, the RMSD values range from a minimum of 0.497 Å to a maximum of 2.090 Å, with an **average RMSD of 1.419 Å** and a **standard deviation of 0.352 Å over 999 frames**. These values indicate moderate conformational flexibility while preserving the overall fold of the protein. The absence of a systematic upward drift in RMSD over time suggests that KRAS remains structurally stable throughout the production run, without evidence of large-scale unfolding or irreversible

conformational changes. The observed fluctuations are consistent with intrinsic backbone motions of the protein under physiological simulation conditions.

d)

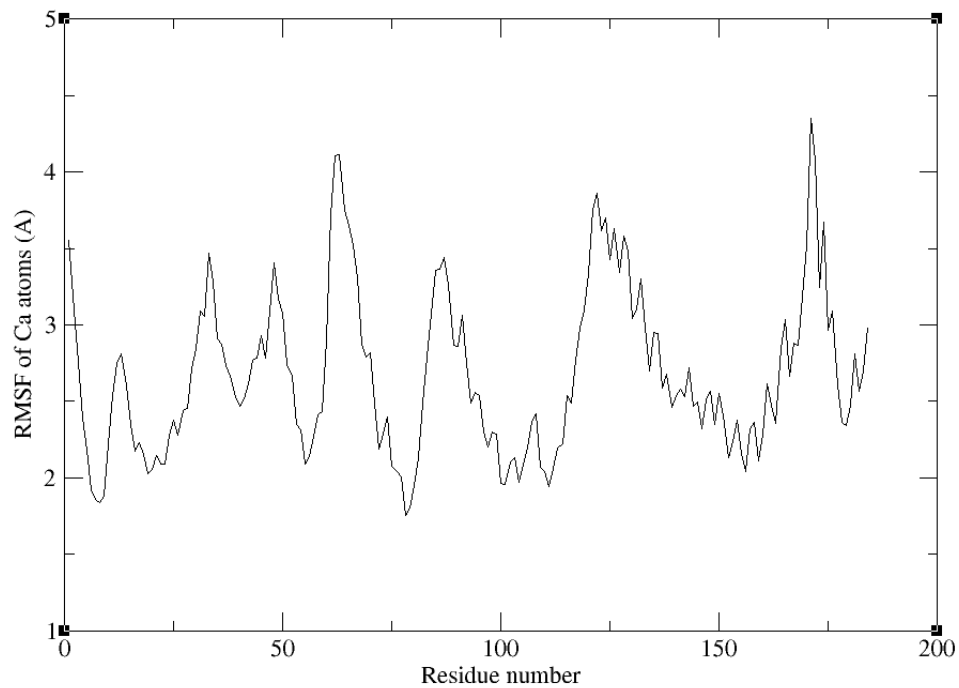


Figure 6d (i). Root-mean-square fluctuation (RMSF) of KRAS C α atoms per residue over the production trajectory, computed after alignment to the reference structure. Higher RMSF values indicate residues with greater conformational mobility (flexibility), while lower values correspond to more rigid, structurally constrained regions.

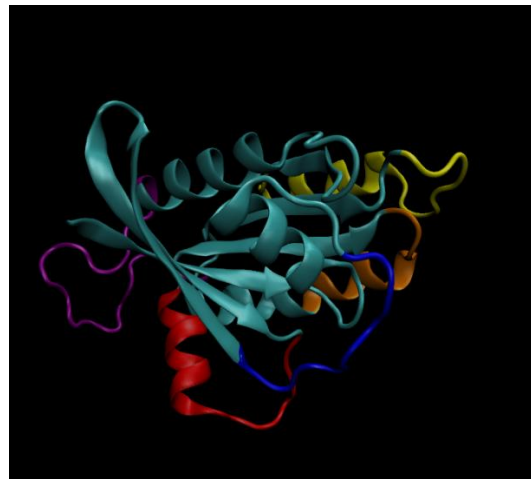


Figure 6d (ii). Structural representation of KRAS with flexible regions highlighted according to the per-residue C α RMSF analysis. The protein core (shown in cyan/green) corresponds to structurally stable α -helical and β -strand elements with low RMSF values, whereas regions with elevated flexibility are colored as follows: Switch I (residues ~30–38, blue), Switch II (residues ~60–75, red), loop region ~85–95 (orange), central flexible segment ~120–140 (yellow), and the highly flexible HVR region ~165–185 (purple). These flexible, solvent-exposed segments exhibit increased conformational mobility and are associated with the functional dynamics of KRAS.

Based on the **per-residue C α** RMSF profile and its mapping onto the KRAS structure, the most flexible parts of the molecule correspond to specific, functional regions rather than the structured core. The highest RMSF values (reaching approximately 4–4.5 Å) are observed in the following segments:

- **Switch I (residues ~30–38, colored blue)**
- **Switch II (residues ~60–75, colored red)**
- **Loop region ~85–95 (colored orange)**
- **Central flexible segment ~120–140 (colored yellow)**
- **HVR region ~165–185 (colored purple), which exhibits the largest RMSF peak**

These regions are predominantly loop-like, surface-exposed, or terminal segments that are not tightly constrained by stable secondary structure or dense packing interactions. In particular, Switch I and Switch II are known from the literature to be intrinsically dynamic motifs that undergo conformational rearrangements during GTP/GDP cycling and during interactions with effectors (e.g., RAF, PI3K) and regulatory proteins (GEFs and GAPs). Their elevated RMSF values therefore reflect functionally required flexibility.

The regions around residues 85–95 and 120–140 also display increased mobility, consistent with their solvent exposure (particularly residues 120–140) and the limited stabilization by regular secondary structure in these segments. The Hypervariable Region (HVR) at the C-terminus (residues ~165–185) shows the highest flexibility, as reflected by the largest RMSF values. This behavior is expected for the intrinsically disordered and lipid-modified HVR, which requires high conformational adaptability to mediate **membrane association** and proper spatial orientation of KRAS.

In contrast, the structured α/β core of KRAS (shown in green) maintains consistently lower RMSF values (≈ 2 Å), indicating reduced mobility due to stabilization by regular secondary structure and extensive intramolecular contacts.

Overall, the RMSF analysis demonstrates that KRAS flexibility is localized primarily in the Switch regions, surface loops, and the HVR, reflecting their functional roles and solvent exposure, while the protein core remains comparatively rigid and structurally stable.

